

Three Sources of Observed Entropy in Reversible Hidden Dynamics: Mixing, Projection Multiplicity, and Inaccessible Degrees of Freedom

Andrew H. Bond*

San José State University, San José, California, USA

(Dated: August 4, 2026)

A previous paper showed that folds of a singular projection create observed entropy with zero hidden information loss. This paper completes that program with three measured experiments that say where observed entropy comes from and when it becomes real. The first experiment is a matched design in which an integrable and a chaotic dynamics share every other parameter while a folding observer and a monotone observer read the same trajectories. Sustained growth of the observed ensemble entropy follows the dynamics under either observer, the multiplicity of hidden states behind one observation follows the observer under either dynamics, and the two mechanisms are independent. The second experiment measures a hierarchy of consumers reading a double fold with increasing access. Their exact conditional entropies form a lawful decreasing chain whose gaps price each additional observable, and the chain is the classical counterpart of a quantum consumer hierarchy measured earlier in the campaign. The third experiment couples the earlier reversible fold cycle to a thermal bath the observer cannot reverse. Reversal restricted to the system fails in proportion to the coupling, reversal extended to the bath succeeds exactly at every coupling, and the record of the leak is invisible to the coarse bath observables measured here. Within this model class, observed entropy growth is mixing, observed ambiguity is the observer's own singularities, and irreversibility is an accounting of which degrees of freedom the observer can reach. Nothing in this paper is a claim about physical spacetime or about the thermodynamics of any real system.

I. INTRODUCTION

An observer watching a hidden dynamical system through a projection can see its recorded entropy change for reasons that have nothing to do with each other. The hidden flow can genuinely mix, spreading an ensemble across the observer's bins. The projection can fold, so that a single observed instant sits over several hidden states at once. And information can leak into degrees of freedom the observer cannot manipulate, so that a reversal protocol available to the observer no longer undoes the evolution. These three sources are routinely described with the same word, entropy, and a central discipline of the present campaign is to refuse that conflation until an experiment forces it or forbids it.

The first paper of this track [1] studied the fold itself. It proved that a nondegenerate fold changes the number of hidden preimages by two while conserving an orientation-weighted count, computed exact branch entropies for canonical control maps, measured the resolution scaling of the caustic, and demonstrated a reversible Hamiltonian ensemble whose observed entropy swings by more than two bits and then returns every bit, with the initial hidden ensemble recovered to near machine precision. That paper established that folds create observed entropy without hidden loss. It deliberately left three questions open. Whether fold multiplicity or hidden mixing controls the growth of observed entropy in a system that has both. What a hierarchy of observers with graded access is worth in exact conditional bits. And when the recoverable

ambiguity of a projection becomes irreversible entropy production.

This paper answers all three within the same model family, and the track is thereby complete. The answers have a common shape. Each candidate source of observed entropy is isolated by a matched design in which the other sources are held fixed, and each answer is a measured number with a falsification bar that was written down before the run. The mixing experiment finds that growth belongs to the dynamics and multiplicity belongs to the observer, with no leakage between the two. The hierarchy experiment finds a lawful chain of exact conditional entropies whose gaps price orientation and branch knowledge separately. The bath experiment finds that irreversibility is not a property of the dynamics at all in this class. It is a property of which momenta the observer is permitted to flip, and the leaked record hides in correlations that no coarse bath observable measured here can see.

The scope of these claims is narrow and stated once. Every result is a computation or a measured simulation in a declared toy model class, graded by the campaign's evidence scheme as demonstrated in model. No claim is made about physical spacetime, about the thermodynamics of any real system, or about quantum gravity.

II. THE MODEL FAMILY AND THE COMPLETED TRACK

All three experiments use the campaign's standard hidden system, two coupled degrees of freedom with Hamiltonian

$$H = \frac{1}{2}(p_t^2 + p_u^2) + \frac{1}{2}\omega_t^2 t^2 + \frac{1}{2}\omega_u^2 u^2 + \frac{\lambda}{4}u^4 + gEtu, \quad (1)$$

* andrew.bond@sjsu.edu; ORCID: 0009-0003-2599-6158

with $\omega_u = 1.2$ and $g = 0.25$ throughout, integrated by time-symmetric methods whose energy drift is monitored in every run. The observed coordinate is t and the observer's records are binned at a declared resolution ϵ . The projection of interest sends the hidden evolution parameter τ to an observed time functional, and the observer's singularities are the critical points of that functional.

The track now comprises six experiments. The fold theorem and the exact branch entropies, the caustic resolution scaling, and the reversible fold cycle are published [1]. The three completions reported here are the mixing comparison, the consumer hierarchy, and the bath coupling. Every result-bearing run writes an append-only evidence record with hashes, and Sec. VII lists them.

III. MIXING VERSUS MULTIPLICITY

A. A matched two-by-two

The design question is whether the growth of observed entropy is controlled by hidden chaotic mixing or by the multiplicity of the observer's projection. A four-system comparison invites leakage, because a system built to mix usually also folds differently. The experiment instead crosses two dynamics with two observers so that neither axis can contaminate the other.

The two dynamics share every parameter of Eq. (1) with $\omega_t = 1$ and $E = 2$ and differ only in the quartic coefficient, $\lambda = 0$ for the integrable member and $\lambda = 0.35$ for the chaotic member. The two observers read the same trajectories through two time functionals, the folding functional $T = t$ and the monotone functional $T = t + 3\tau$. The monotone functional never folds because $dT/d\tau = p_t + 3$ and $|p_t|$ stays well below 3 in this family. The decisive property of the design is that the two functionals differ only by a linear term in τ , so at every instant the ensemble marginal of t is the same for both observers. Growth of the binned ensemble entropy therefore cannot depend on the observer, and the multiplicity of worldline states behind one observed level cannot depend on anything but the observer. The two mechanisms are separated by construction, and the experiment measures whether each one actually moves.

A 20000-member ensemble with thermal hidden initial conditions is integrated to $\tau = 40$ for each dynamics, and the centered binned entropy of the observed marginal is recorded along the way at $\epsilon = 0.05$. On a reference trajectory the unsigned crossing count of each observed level, the multiplicity a worldline observer must resolve, is measured for both functionals.

B. Result

Figure 1 shows the two entropy curves. The total rise over the run is nearly identical, 5.71 bits for the integrable dynamics and 5.83 bits for the chaotic one, because the

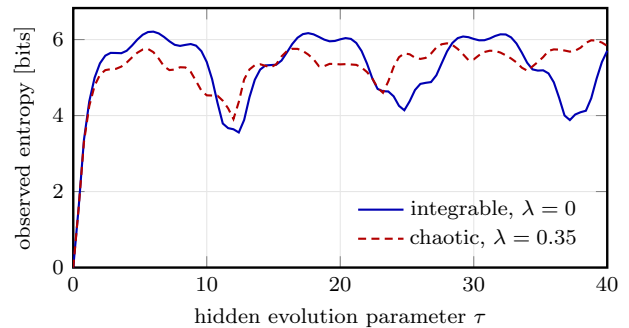


FIG. 1. Observed binned entropy of the t marginal for the two matched dynamics. The early rise is common linear dispersion and does not separate the systems. The late behavior does. Over the second half of the run the integrable curve recurs with fitted slope -0.024 bits per unit τ while the chaotic curve continues to grow at $+0.030$ bits per unit τ . Data from the committed evidence record of the mixing experiment.

early spread of the ensemble is ordinary linear dispersion and is common to both. An honest reading must therefore refuse the total rise as a discriminator. The late-time slope discriminates. Fitted over the second half of the run, the integrable curve turns over and recurs at -0.024 bits per unit τ while the chaotic curve keeps growing at $+0.030$ bits per unit τ . Sustained growth follows the dynamics, under either observer, because the observers share the marginal by construction.

Multiplicity behaves in the complementary way. The folding observer sees a mean unsigned crossing multiplicity between 9.6 and 10.2 per observed level, with a maximum of 12, under both dynamics. The monotone observer sees exactly one crossing per level everywhere, under both dynamics. Multiplicity follows the observer's singularities and is indifferent to whether the hidden flow mixes.

The falsification bar, fixed before the run, was met. A chaotic late slope at or below the integrable one, a folding observer with mean multiplicity at or below 1.5, or a monotone observer seeing any level more than once would each have invalidated the reading. The conclusion is that mixing and multiplicity are independent axes. Mixing pumps observed entropy. Folds create observed ambiguity. Neither implies the other.

IV. THE CONSUMER HIERARCHY

A. Graded access to a double fold

The second completion prices what each additional observable is worth to an observer of a singular projection. The system is the double fold $t = \tau^3 - \tau$ on the declared interval $|\tau| \leq 1.6$, the map whose three branches carry only two orientations because the outer branches share the sign of $dt/d\tau$. That degeneracy is the reason this map is the right one. A consumer who learns the orientation has not learned the branch, so the rungs of the hierarchy

TABLE I. The measured consumer chain on the double fold. Each row states what the consumer reads and the exact conditional entropy of the hidden state given that reading. The residual of the last rung is within-branch positional uncertainty set by the declared grid and bin resolutions.

Consumer reads	$H(\text{hidden} \text{reading})$ [bits]
binned position only	9.5555
position and orientation	8.8307
position and branch label	8.5453
full hidden state	0

are genuinely distinct.

The hidden state is uniform on a τ grid of spacing 10^{-4} and the observed time is binned at $\epsilon = 0.05$. Three consumers read the system. The first sees only the binned observed time. The second sees the binned time and the orientation, the sign of $dt/d\tau$. The third sees the binned time and the full branch label. Because the state space is a finite grid, every conditional entropy is an exact finite sum with no estimator and no sampling error.

B. Result

Table I shows the chain. The ordering $9.5555 \geq 8.8307 \geq 8.5453 \geq 0$ holds with strictly positive gaps at every rung. Orientation is worth 0.7248 bits. The branch label is worth a further 0.2854 bits beyond orientation, and that gap is exactly the price of the outer-branch degeneracy, since orientation resolves the middle branch but cannot split the outer pair. The residual 8.5453 bits of the branch consumer is within-branch positional uncertainty fixed by the grid and bin resolutions, not by the fold. A branch is an interval of hidden states rather than a point, so no discrete label can take the chain to zero.

The campaign’s quantum track measured the same structure in an Ising consumer hierarchy, where coarser consumers pay quantifiable distinguishability costs in a lawful degradation chain [2]. The present chain is the classical counterpart that the quantum paper’s reduction argument cites. The gaps are classical prices in exact bits, and the ordering is forced by the data-processing inequality in its most elementary form.

V. REVERSAL REACH AND ENTROPY PRODUCTION

A. The cycle with a bath attached

The published reversible fold cycle showed that more than two bits of observed entropy can appear and vanish with zero hidden loss, because flipping every momentum retraces the evolution exactly [1]. The third completion asks what changes when the system is coupled to an environment. The hidden coordinate u of Eq. (1), with

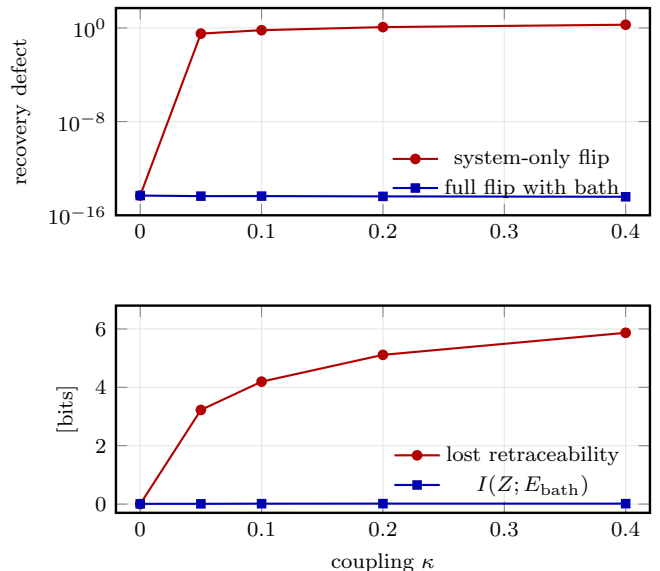


FIG. 2. Top panel. Root-mean-square recovery defect of the hidden system state after the round trip, for the reversal the observer can perform and for the control that also flips the bath. The observer’s reversal fails monotonically with coupling while the full flip recovers to better than 5×10^{-15} at every coupling. Bottom panel. The entropy retrace defect grows to 5.87 bits while the mutual information between the observed coordinate and the bath energy never leaves the estimator floor. Data from the committed evidence record of the bath experiment.

$\omega_t = 1$, $E = 1$, and $\lambda = 0.1$, is coupled to a bath of 32 harmonic modes with frequencies spread uniformly over $[0.5, 2.5]$, thermal initial conditions, and bilinear coupling of strength $\kappa/\sqrt{32}$ per mode in the style of the standard oscillator-bath models [4]. The full evolution remains Hamiltonian and time-symmetric, so in the full space nothing is irreversible at any coupling.

The observer’s limitation is operational. The observer can flip the system momenta but cannot touch the bath. For each coupling $\kappa \in \{0, 0.05, 0.1, 0.2, 0.4\}$ an 8000-member ensemble is integrated forward to $\tau = 20$, and from the same stored turning state two return legs are run. One flips only the system momenta, which is what the observer can do. The other flips the bath momenta as well, which is the control. The witnesses are the hidden-state recovery defect of each return, the retrace defect of the observed entropy curve, and the plugin mutual information between the binned observed coordinate and the binned bath energy.

B. Result

Figure 2 carries both findings. At $\kappa = 0$ the cycle reproduces the published result exactly, with recovery defect 4.8×10^{-15} and entropy retrace defect exactly zero. As the coupling grows, the reversal available to the

observer fails monotonically, with recovery defects 0.33, 0.64, 1.18, and 1.89 across the grid and a retrace failure that reaches 5.87 bits. The control tells the other half of the story. The full flip, including the bath, recovers the hidden state to better than 5×10^{-15} at every coupling. The dynamics never became irreversible. What changed is the reach of the observer’s reversal protocol. In this model class, entropy production is an accounting of which degrees of freedom the observer can flip, and the published claim that observed entropy carries no hidden loss survives intact in the enlarged space.

The second finding was not the expected one, and the runner’s first version asserted its opposite. The natural expectation is that the leaked information should be visible in the bath, for instance as mutual information between the observed coordinate and the bath energy. Measured, it is not. Across the whole grid that mutual information stays between 0.009 and 0.016 bits, within a factor of two of its zero-coupling estimator floor, while the observer’s lost retraceability grows to almost six bits. The mean bath energy is no witness either, since without a counter-term the coupling shifts the equilibrium and the bath energy actually falls at strong coupling. The record of the leak lives in system-bath microstate correlations that none of the coarse bath functionals measured here can see. Recovery therefore requires microstate access, not a macroscopic bath reading. The revised assertion set demands this invisibility, and a coarse mutual information rising above 0.05 bits would now refute the reading. This is the classical face of a familiar quantum lesson, that what an environment records and what a coarse observer of that environment can read are very different things [5].

VI. WHAT THE TRACK ESTABLISHES

Table II registers the claims of the completed track with their labels. Read together, the six experiments give the campaign’s answer to where observed entropy comes from in this model class. The growth of observed entropy is mixing, a property of the hidden dynamics. The ambiguity of an observation is multiplicity, a property of the observer’s own singularities, worth an exactly computable number of bits to resolve. The irreversibility of observed

entropy is neither of these. It is an accounting statement about which degrees of freedom the observer’s reversal can reach, it appears at the moment ambiguity leaks into unreachable ones, and it leaves no record that a coarse reading of the environment can find.

Three cautions bound the reading. These are toy Hamiltonians with a handful of degrees of freedom, and no thermodynamic limit is taken anywhere in this paper. The bath experiment measures one family of coarse bath observables, and a cleverer functional of the bath might see more of the leak than the energy does. And the mixing experiment’s chaotic member was diagnosed by its behavior rather than by a Lyapunov measurement, so the label chaotic is operational. None of these cautions weakens the matched comparisons, which are internal to each experiment.

VII. METHODS AND REPRODUCIBILITY

All instruments derive from the campaign’s sealed tolerance and instrument freezes, registered as PF0-FREEZE-001 and PEQO-FREEZE-002, which fix numerical bars in advance and void themselves if edited. The three experiments of this paper are exploratory-grade instruments validated against exact controls, the zero-coupling limit for the bath cycle, the monotone observer for the multiplicity count, and the exact finite sums for the hierarchy. Every result-bearing run writes an append-only JSON evidence record containing the complete parameter set, seeds, outcome arrays, and a canonical SHA-256 hash of the record body, and the records for the three experiments here are `pe3-mixing.json`, `pe5-hierarchy.json`, and `pe4-noise.json` in the campaign repository. Every number and every figure coordinate in this paper is read from those committed records or recomputed deterministically from committed instrument code by a committed script, with no free parameters. One instrument correction made during this work is recorded in the repository history. The degenerate initial ensemble of the bath cycle initially sat exactly on a bin edge, where round-trip noise at the 10^{-16} level split one bin into two and produced a spurious retrace defect, and the binning grid was shifted by half a bin before any evidence was recorded.

-
- [1] A. H. Bond, Entropy across singular projections: fiber multiplicity, caustics, and observer-relative information, Zenodo (2026), doi:10.5281/zenodo.21789011.
 - [2] A. H. Bond, Consumer-relative quantum distinguishability, Zenodo (2026), doi:10.5281/zenodo.21789046.
 - [3] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley-Interscience, Hoboken, 2006).
 - [4] A. O. Caldeira and A. J. Leggett, Path integral approach to quantum Brownian motion, *Physica A* **121**, 587 (1983).
 - [5] W. H. Zurek, Decoherence, einselection, and the quantum origins of the classical, *Rev. Mod. Phys.* **75**, 715 (2003).
 - [6] R. Landauer, Irreversibility and heat generation in the computing process, *IBM J. Res. Dev.* **5**, 183 (1961).
 - [7] C. H. Bennett, The thermodynamics of computation, a review, *Int. J. Theor. Phys.* **21**, 905 (1982).
 - [8] V. I. Arnold, *Catastrophe Theory*, 3rd ed. (Springer-Verlag, Berlin, 1992).

TABLE II. Claim register for the completed entropy track. Labels follow the campaign’s evidence scheme. Proved means a theorem with a written proof. Exact means a closed-form computation verified in independent implementations. Demonstrated in model means a measured simulation with declared parameters, append-only evidence records, and falsification bars fixed before the run.

Experiment	Claim	Label
Fold theorem [1]	a nondegenerate fold changes fiber multiplicity by two and conserves the signed count	proved
Branch entropies [1]	fold one bit, symmetric double fold one and one half bits, degenerate monotone map zero	exact
Caustic scaling [1]	binned caustic entropy converges to its differential limit as the square root of the bin width	demonstrated in model
Reversible cycle [1]	observed entropy swings above two bits with zero hidden loss and exact retrace	demonstrated in model
Mixing versus multiplicity (this paper)	sustained entropy growth follows the dynamics, multiplicity follows the observer, independently	demonstrated in model
Consumer hierarchy (this paper)	exact conditional chain with positive gaps pricing orientation and branch knowledge separately	demonstrated in model
Bath coupling (this paper)	reversal failure is set by the observer’s reach, the full flip recovers exactly at every coupling, and the leak is invisible to coarse bath observables	demonstrated in model